A Brief Introduction to Phylogenetic Causal Inference

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EEB 7500 01 (SP26): Phylogenetic Comparative Methods

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Why Causal Inference?

Disclaimers:

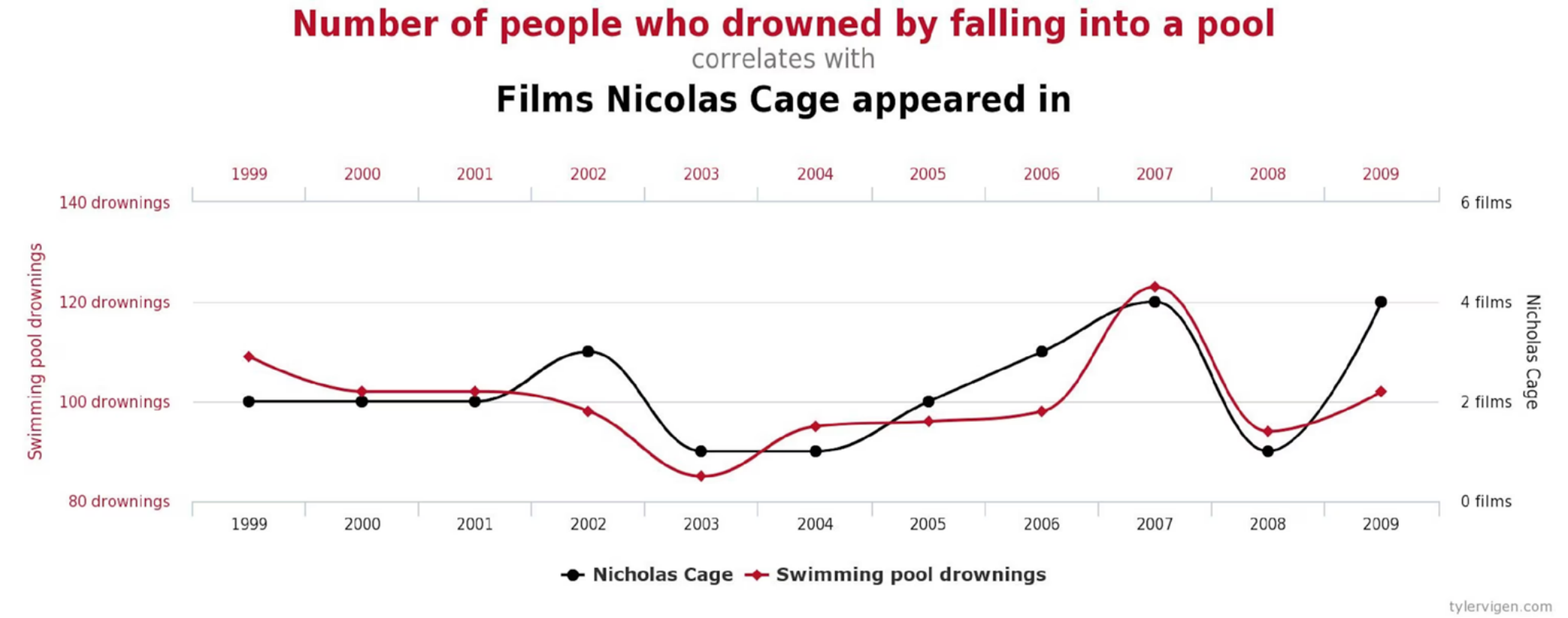
- Using the term in macroevolution will probably still get you in trouble
- It will not save you from “correlation ≠ causation”
- This will not fully equip you to solve your own causal problems today
- I am not presenting software/method
- CI is a logical framework not a single tool

Goals:

- Make assumptions & inference strategies explicit
- Clarify how CI differs from traditional practice
- Align familiar comparative methods with causal ideas
- Remind that thinking causally helps generate testable predictions

Correlation does not equal causation

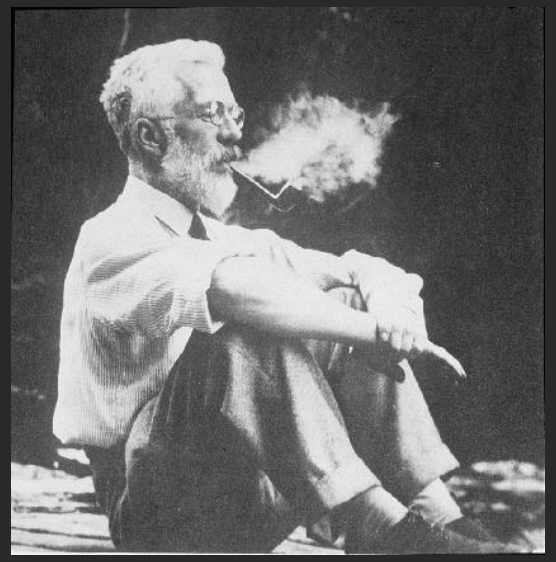
Still true.



Causes in Science

What is the ‘gold standard’ for supporting causal claims in science?

Experiments:
Adding the “do” to science



RA Fisher

Macroevolution = *(mostly) observational science*

Do we care about/talk about causes in macroevolution?

"ecologists [macroevolutionary biologists] often qualify their results verbally to avoid making causal claims—even when their research focus is causal understanding, rather than description…This practice muddies the waters and can create confusion over whether an author is claiming an association or implying causation while allowing themselves plausible deniability."

-Byrnes & Dee, 2024

Do we care about/talk about causes in macroevolution?

Google Scholar "causes of diversification" "phylogeny"

Articles About 365 results (**0.08** sec) 365 results

Google Scholar "drivers of diversification" "phylogeny"

Articles About 1,570 results (**0.08** sec) 1,570 results

...yes, but we may hide the claims (and assumptions) behind weaker language

What is a cause?

Which variable 'listens' to which?

→ If we intervene on one variable, will the other respond?
→ This is different from correlation/association!

Example: Breeder's Equation

$$R = h^2 s$$

Evolutionary **[R]**esponse to selection = **[h²]**eritability * **[s]**trength of natural selection

Mathematical rules: rearranging allowed

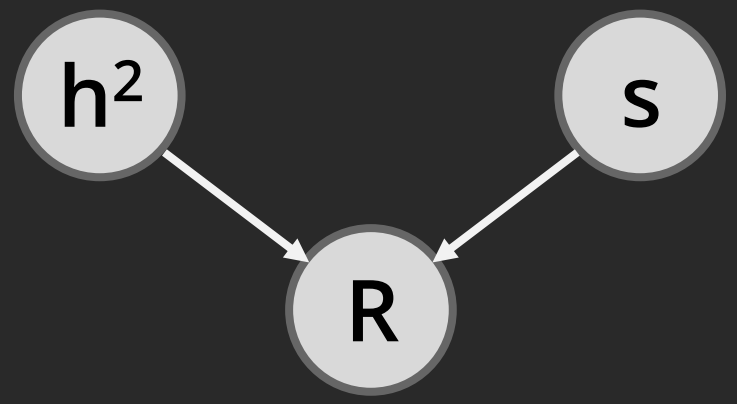
$$R/h^2 = s$$

Causally: Can increasing h^2 decrease s?

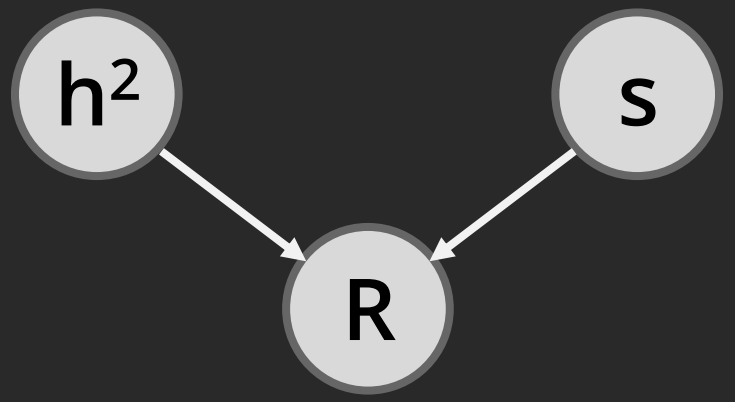
No!

"Do" operator: Causal effect is **p(Y | *do*(X = x)) ≠ p(Y | *do*(X = x'))**

1. ***h²*** and ***s*** are causes of R
2. ***h²*** is not a cause of ***s*** (and vice versa)
3. Manipulating ***R*** doesn't go back in time and affect original ***h²*** & ***s***...



Directed Acyclic Graph as Causal Models



"Do" operator: Causal effect is **p(Y | *do*(X = x)) ≠ p(Y | *do*(X = x'))**

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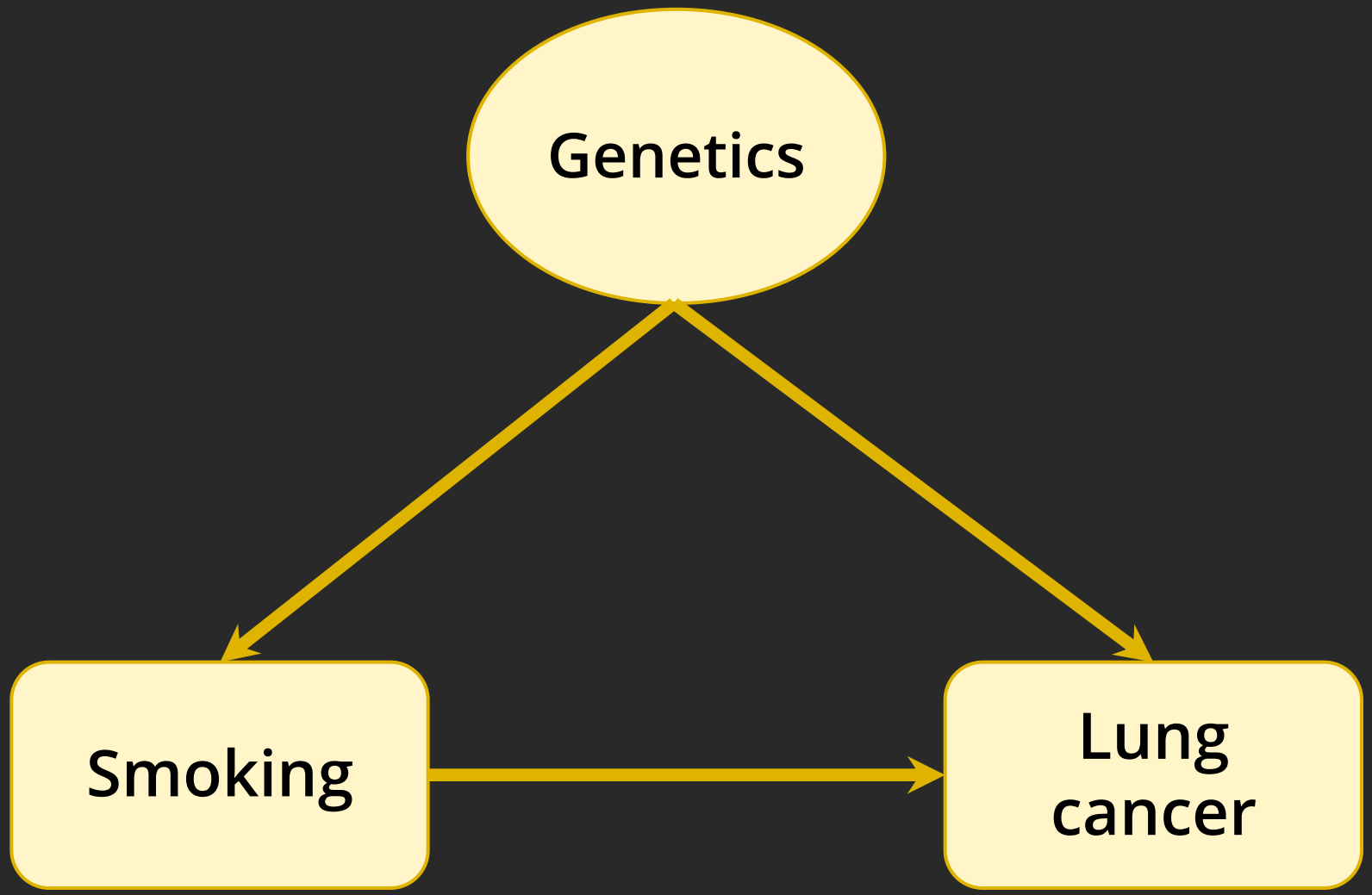
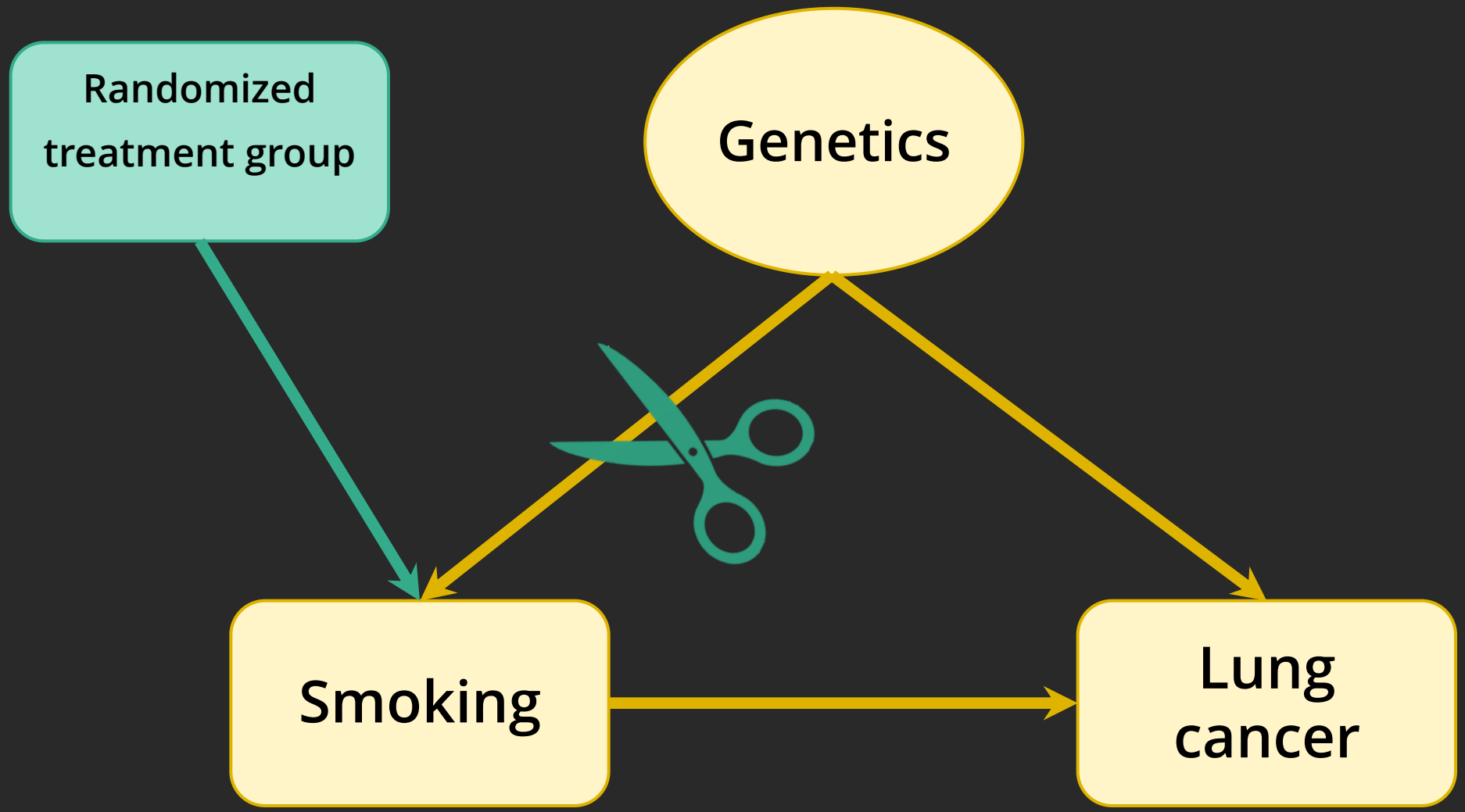
Confounding



Confounding via Experiments

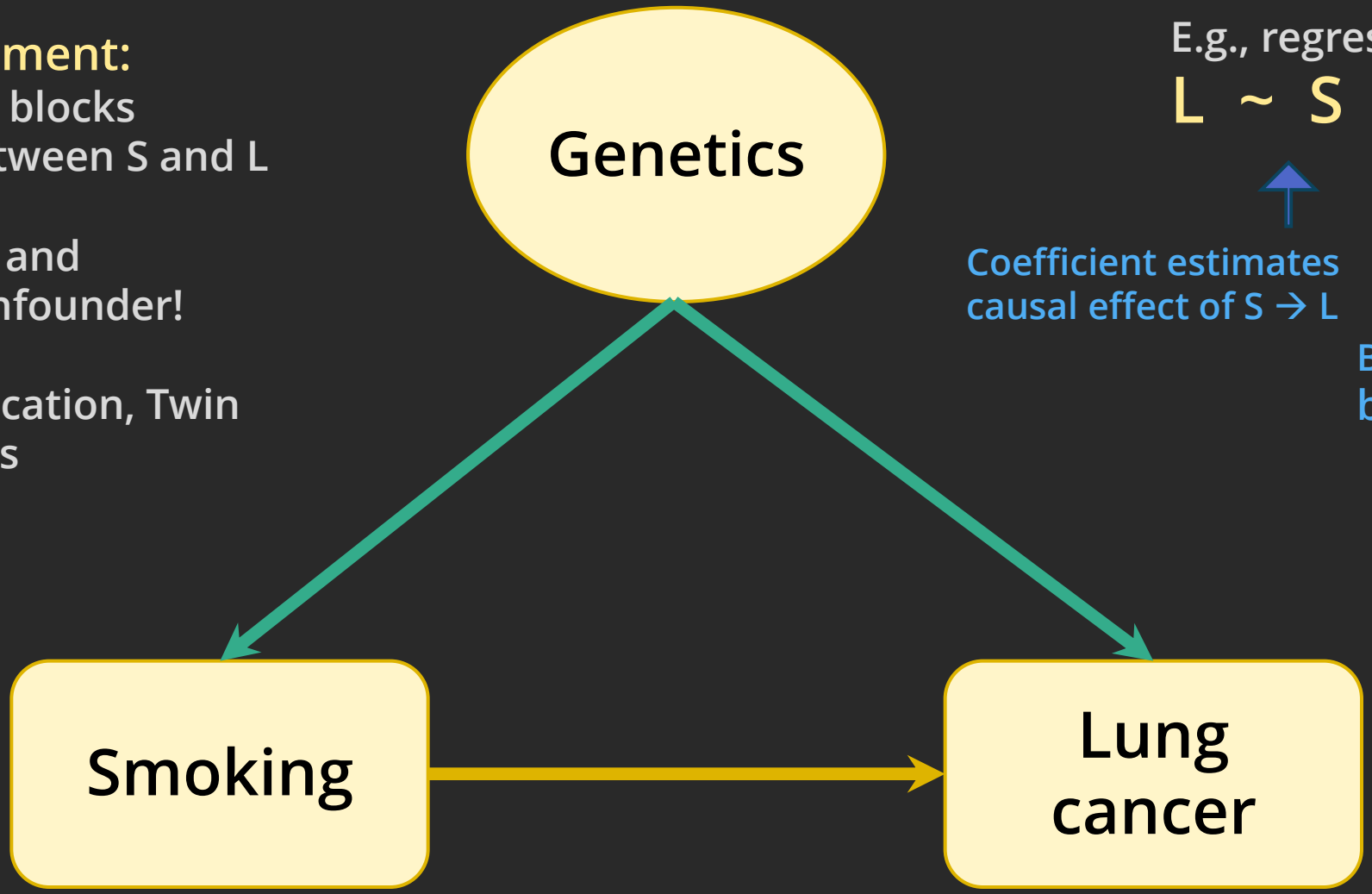


Confounding via Conditioning

Backdoor adjustment:
Conditioning on G blocks backdoor path between S and L

Requires knowing and measuring the confounder!

Population stratification, Twin studies, Genotypes etc.



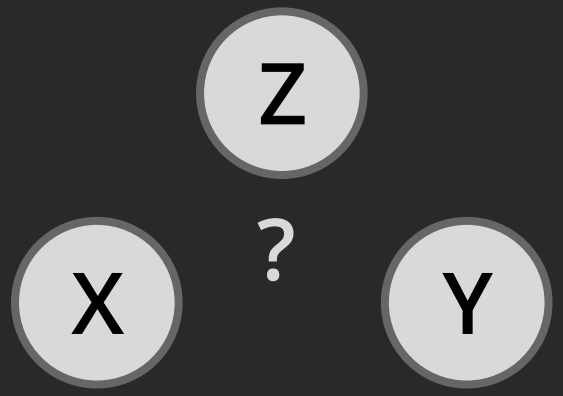
E.g., regression

L ~ S + G

Coefficient estimates causal effect of S → L

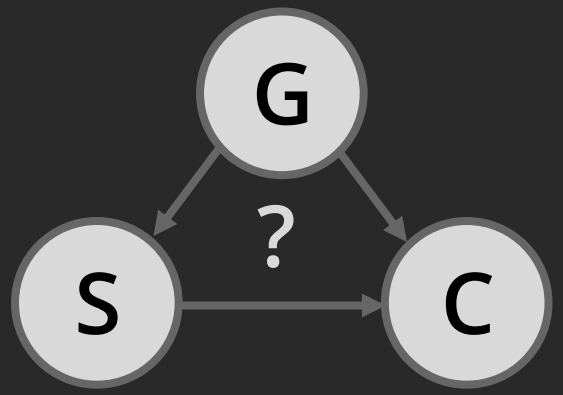
Blocks non-causal backdoor b/w S & L

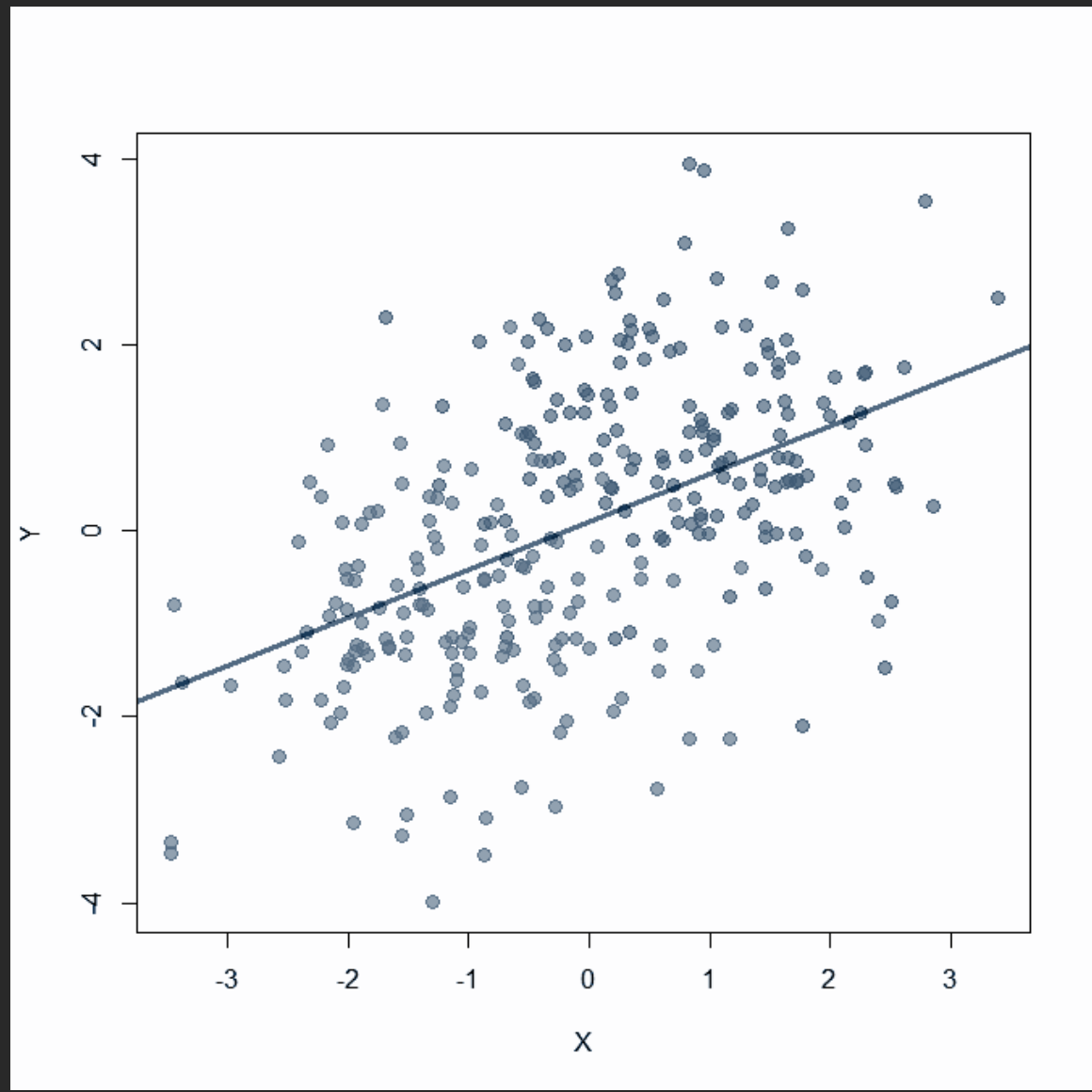
The “Elemental Confounds”



- Basic ways of how three variables can be causally connected
- Rules for how information flows reveal confounding dangers
- Larger DAGs can be deconstructed into these!

The “Elemental Confounds”



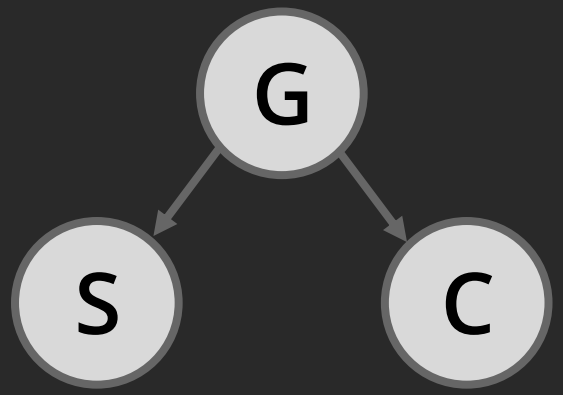


Example:

- Smoking (S) is correlated with cancer (C)
- We wonder whether genetics (G) could affect both

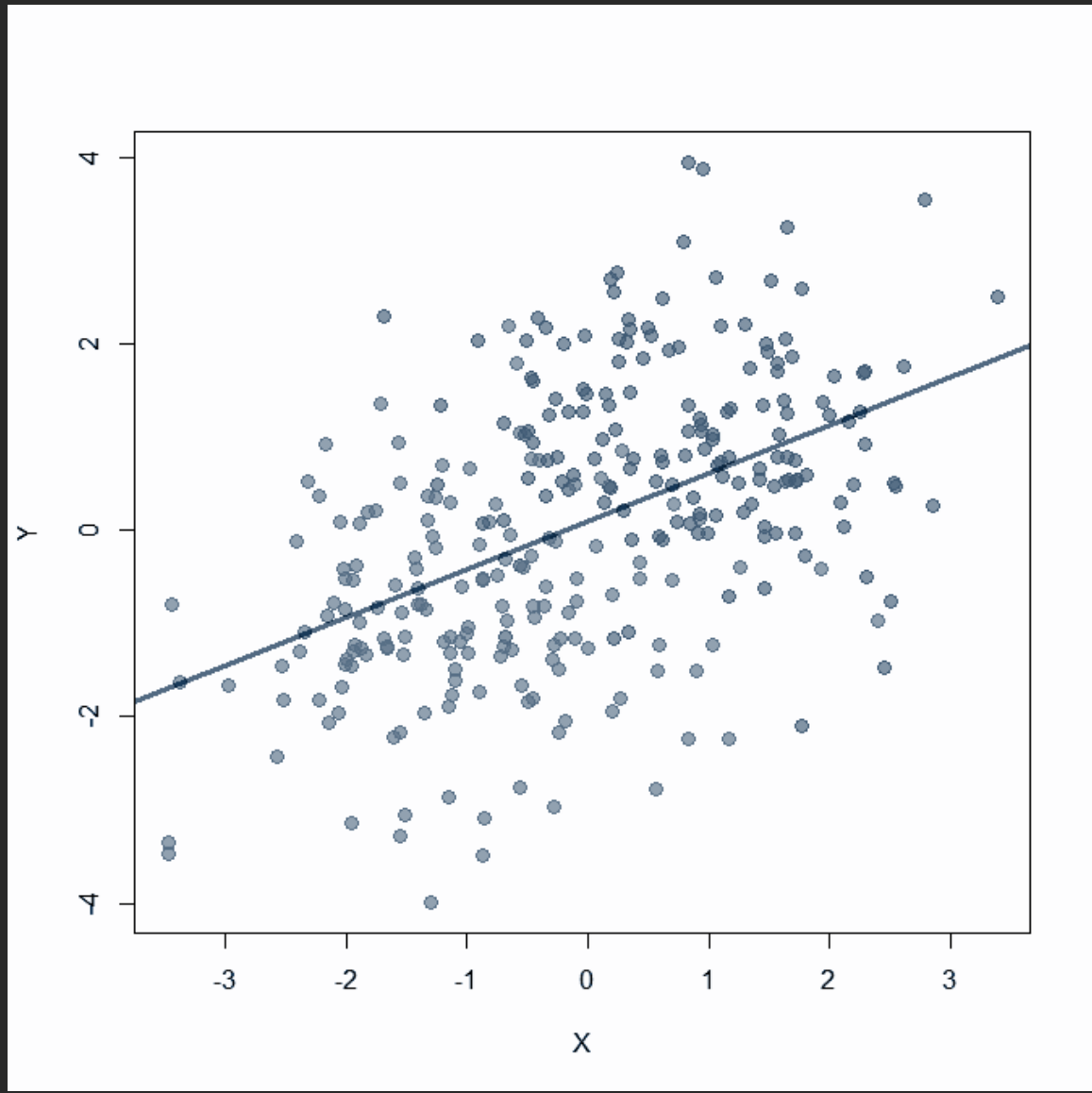
→ **Should we include G in our regression?**

Fork: common cause as confounder

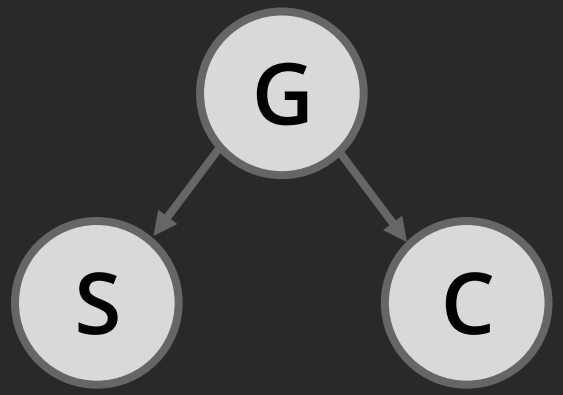



```
lm(formula = C ~ S)

            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.09541    0.07345   1.299    0.195
S            0.51315    0.05350   9.592   <2e-16 ***
```

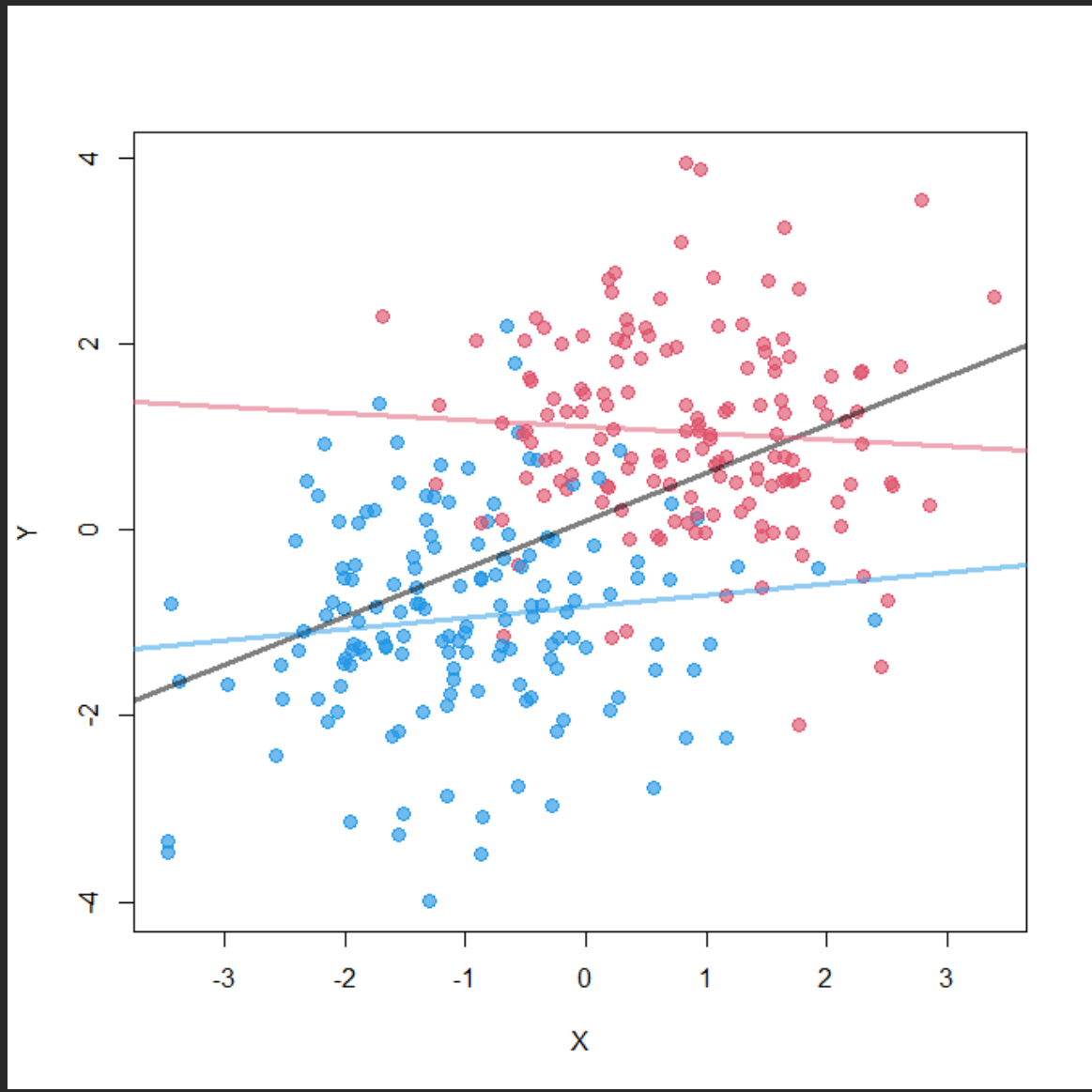


Fork: common cause as confounder




```
lm(formula = C ~ S + G)

            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.91314    0.10648  -8.576 5.53e-16 ***
S            0.03208    0.06091   0.527    0.599
G            1.93154    0.16702  11.565  < 2e-16 ***
```



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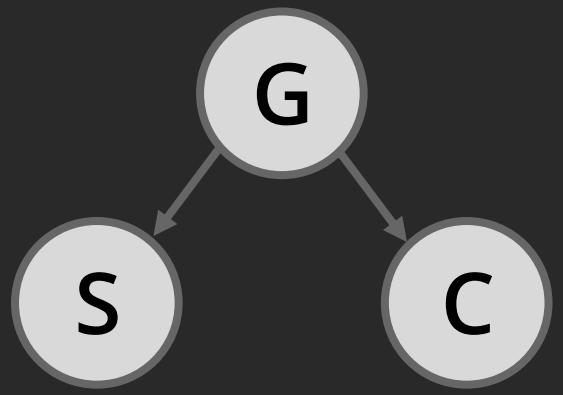
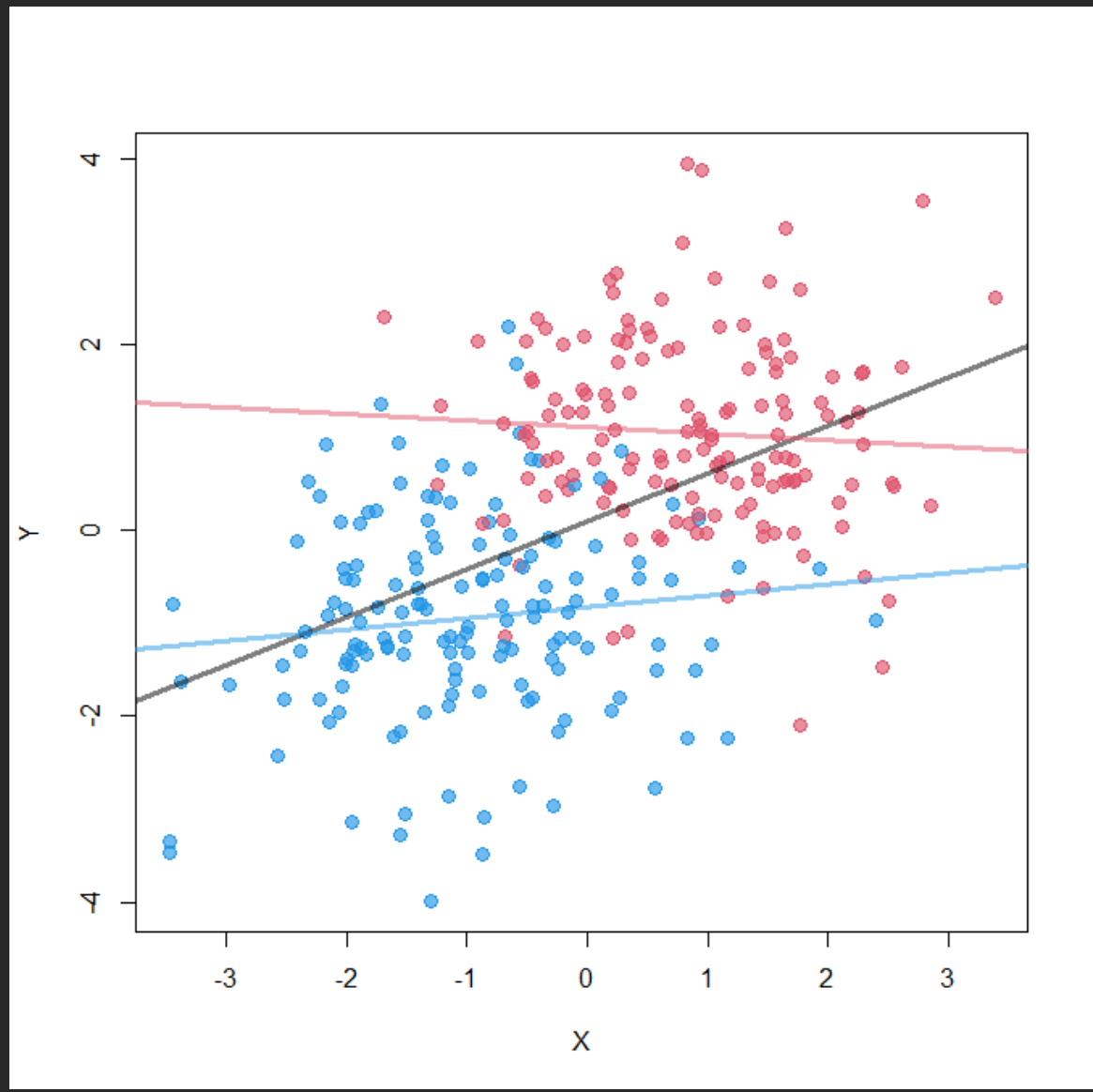
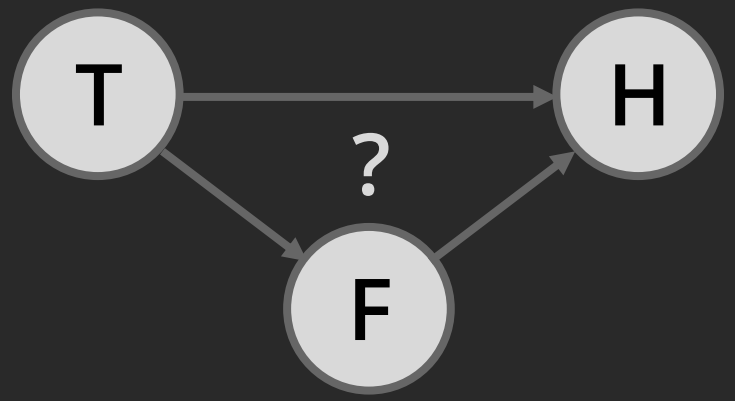


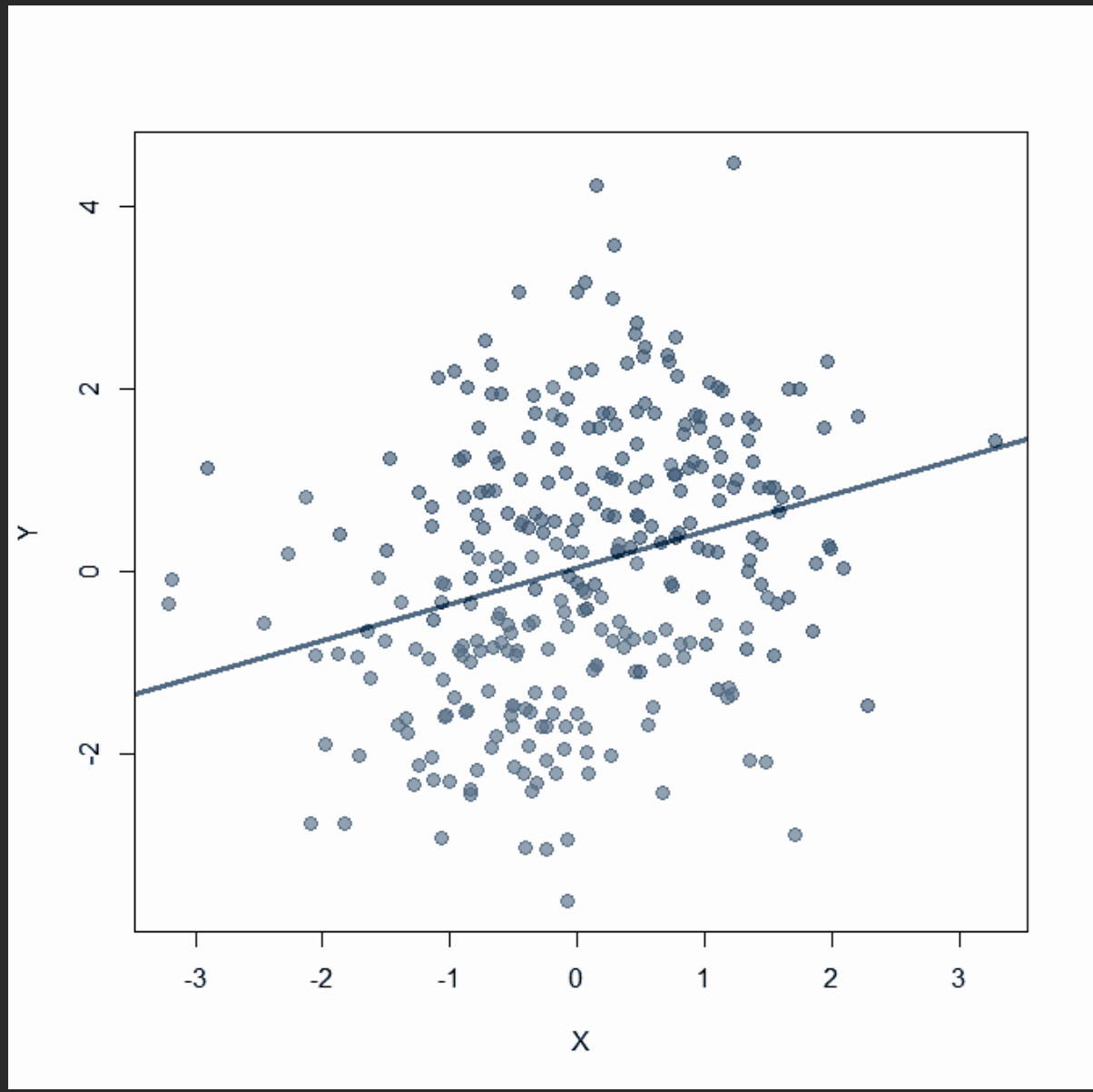
Missing confounder creates spurious correlation!

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.91314    0.10648  -8.576 5.53e-16 ***
S            0.03208    0.06091   0.527    0.599
G            1.93154    0.16702  11.565  < 2e-16 ***
```



The “Elemental Confounds”





Example:

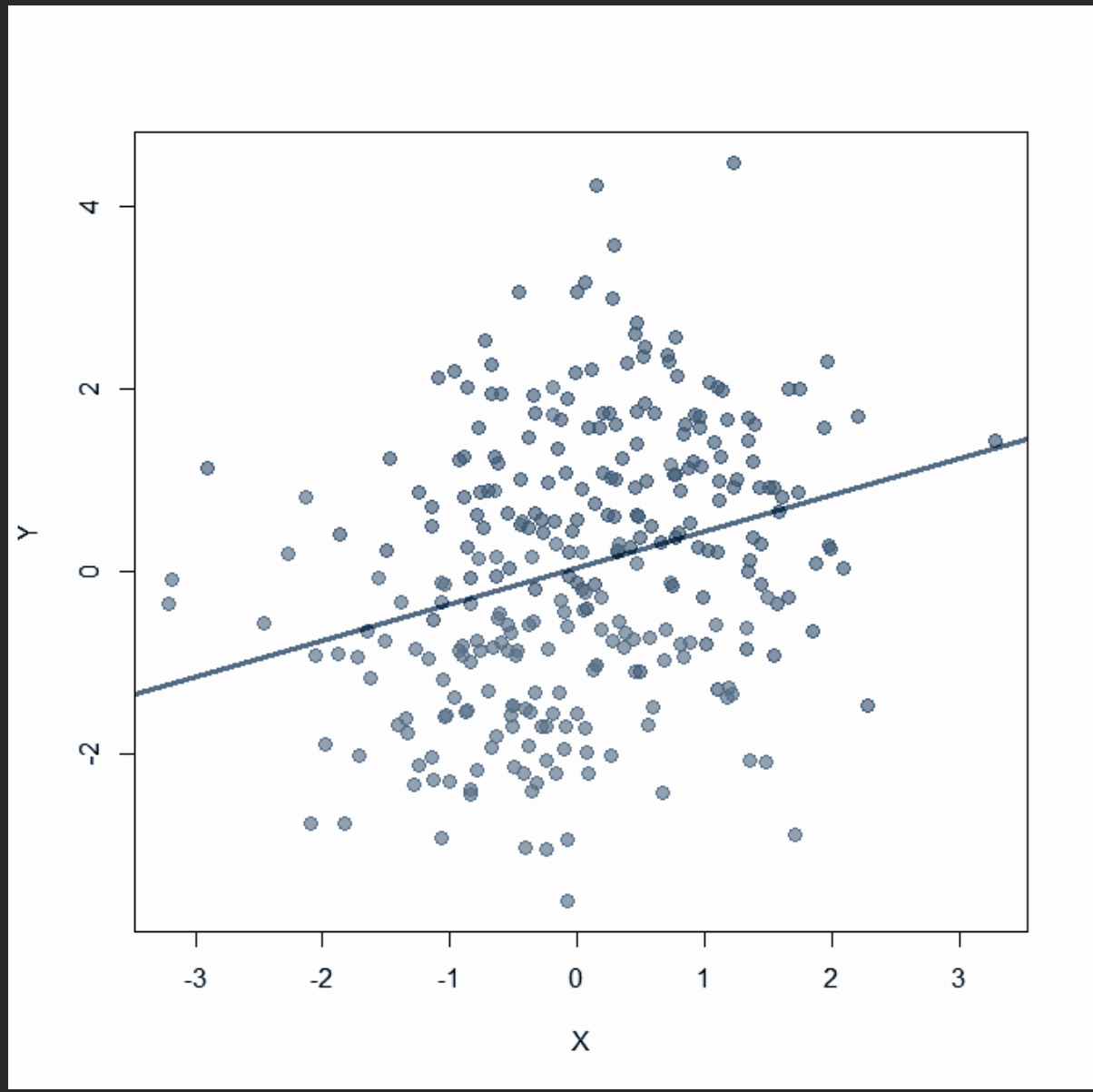
- Fungus (F) affects plant height (H)
- Experiment to test whether treatment (T) works

→ **Should we include F in our regression?**

Pipe: mediator

```
lm(formula = H ~ T)

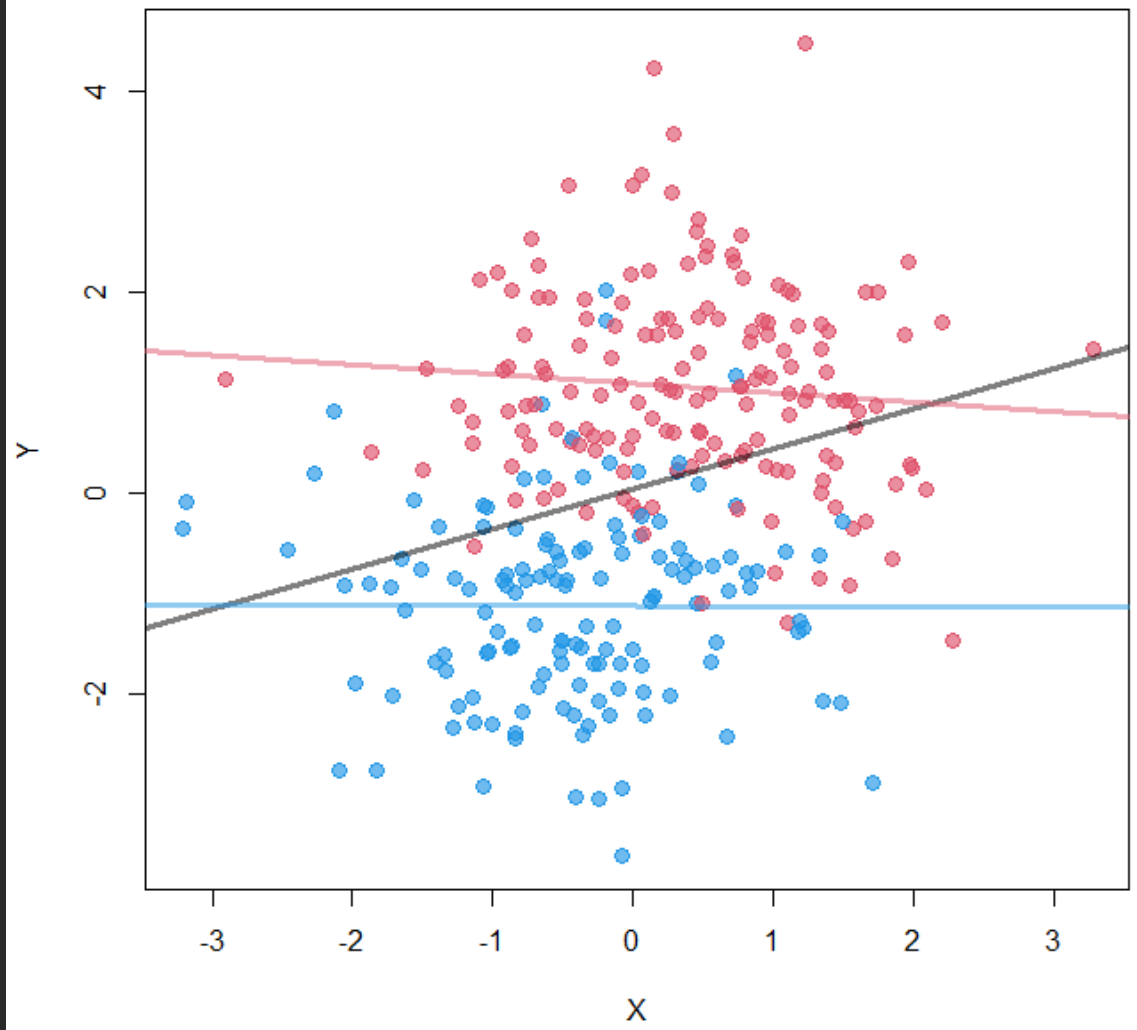
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.04652    0.08174   0.569     0.57
T            0.39762    0.08046   4.942 1.29e-06 ***
```



Pipe: mediator

```
lm(formula = H ~ T + F)

            Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.13684    0.08815 -12.897   <2e-16 ***
T           -0.05336    0.06185  -0.863    0.389
F            2.22261    0.12596  17.645   <2e-16 ***
```



Pipe: mediator

Blocking effects
(Post-treatment bias)

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.13684    0.08815 -12.897   <2e-16 ***
T           -0.05336    0.06185  -0.863    0.389
F            2.22261    0.12596  17.645   <2e-16 ***
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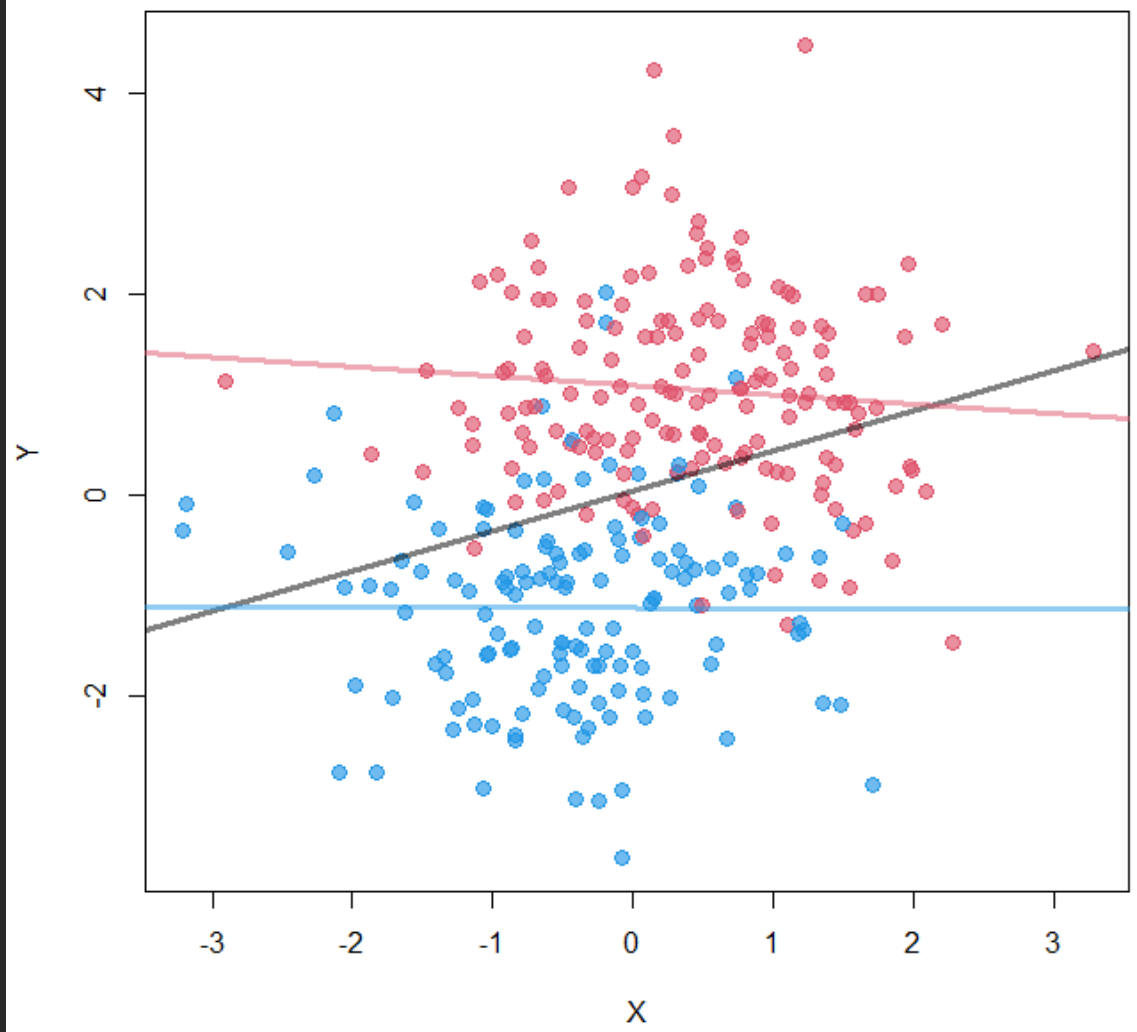

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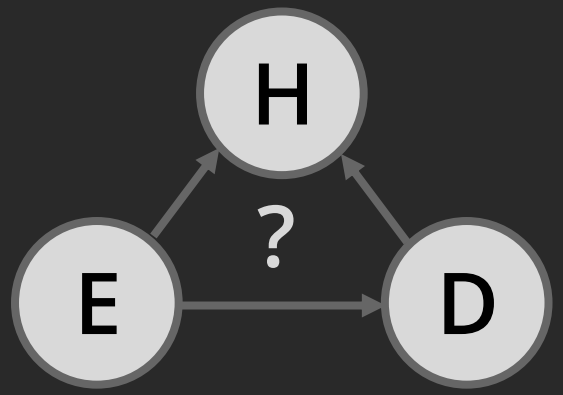
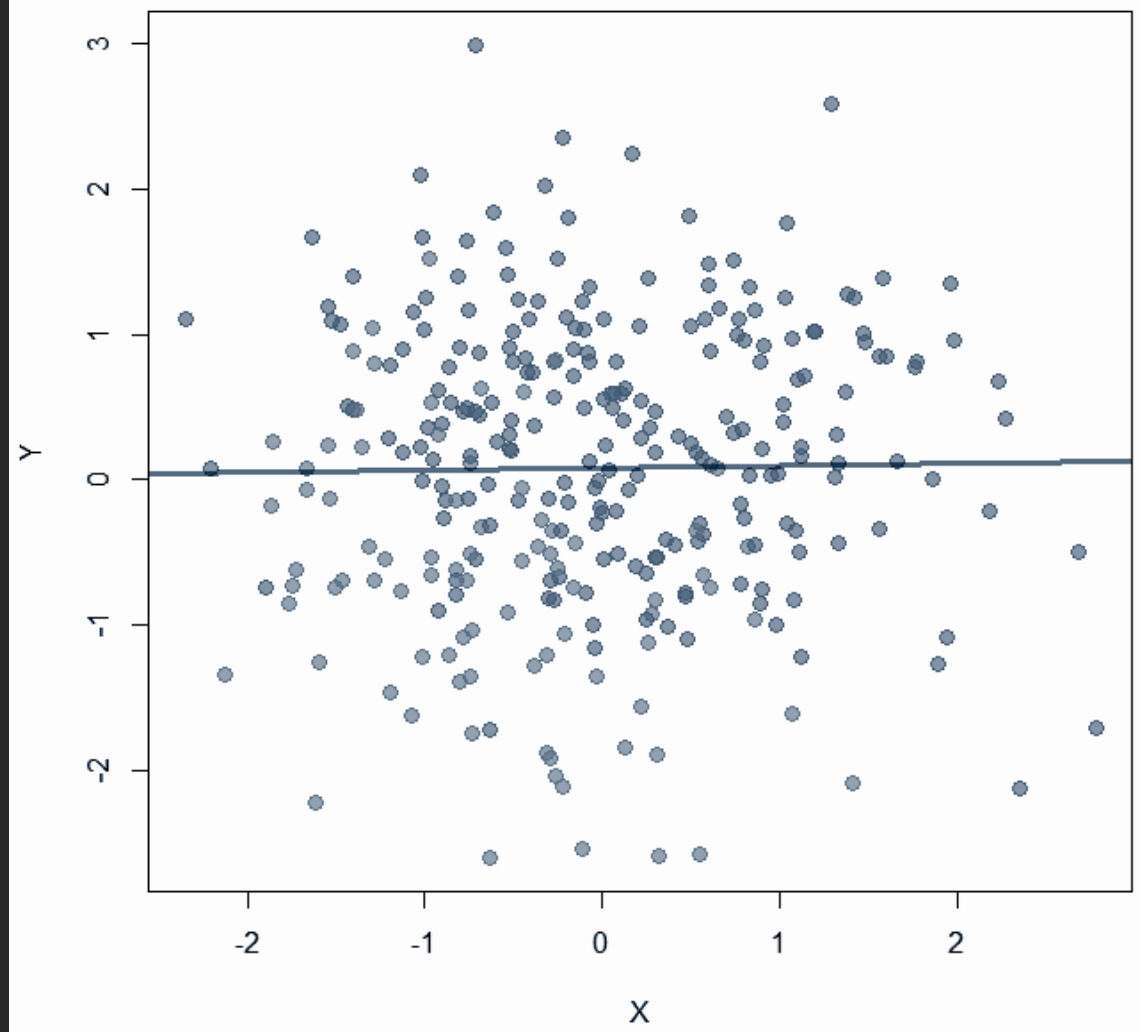


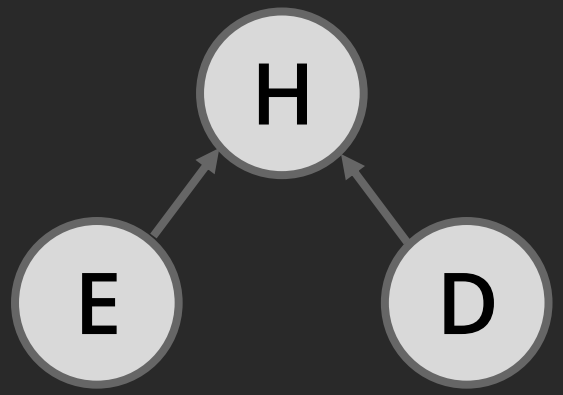


Example:

- Studying health effects of exercise (E) on diet (D)
- Hospitalisation (H) as related proxy for health

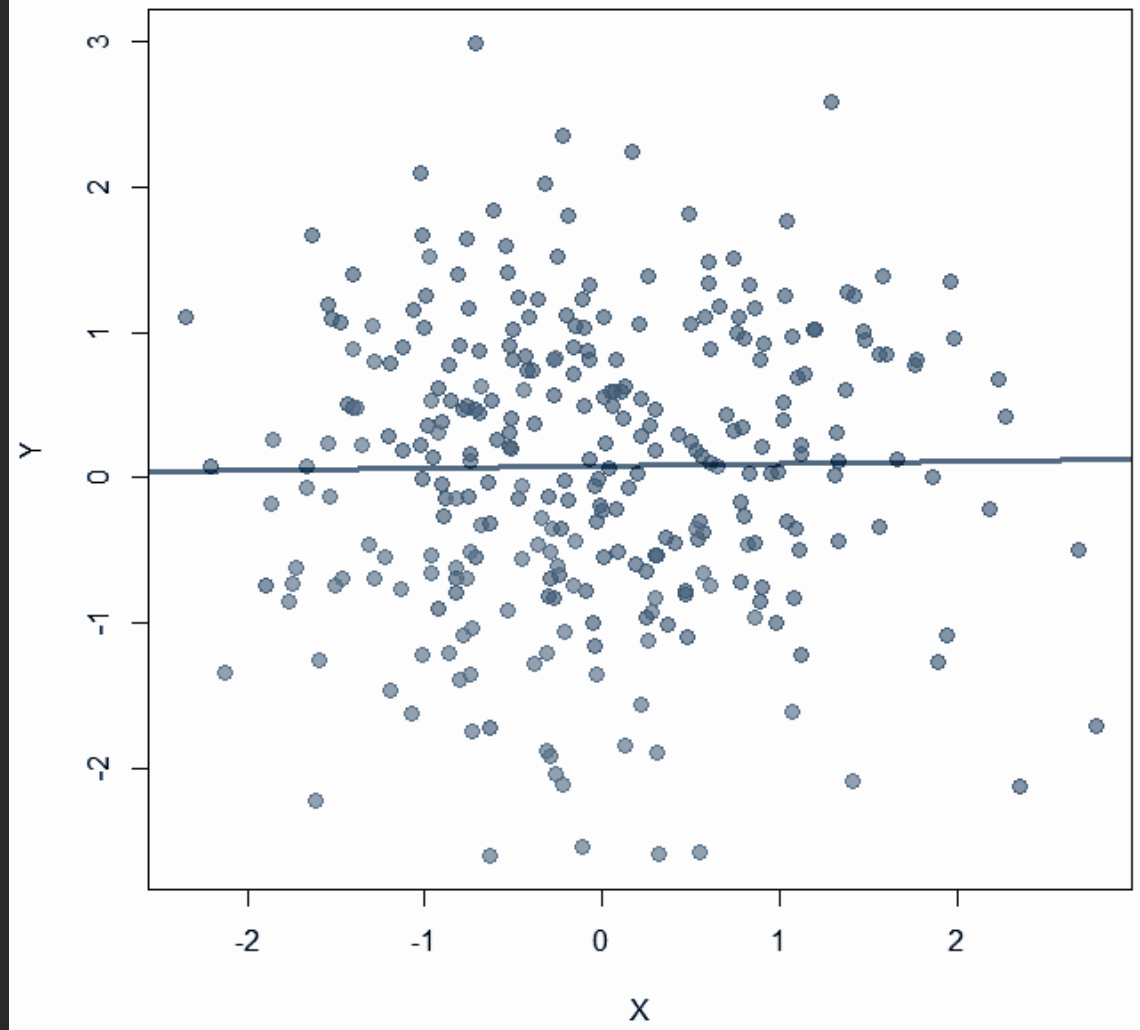
→ **Should we include H in our regression?**

Collider: common effect




```
lm(formula = D ~ E)

            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.08051    0.05753   1.399    0.163
E            0.01505    0.05947   0.253    0.800
```



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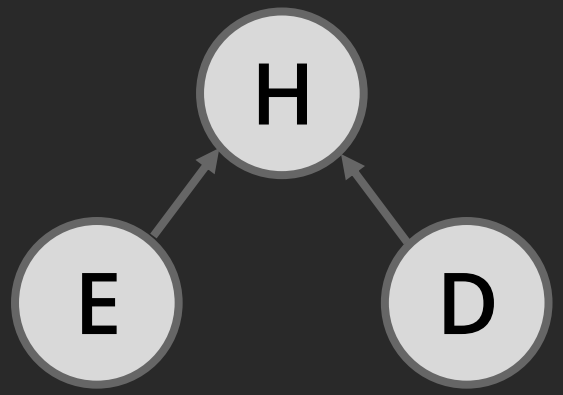
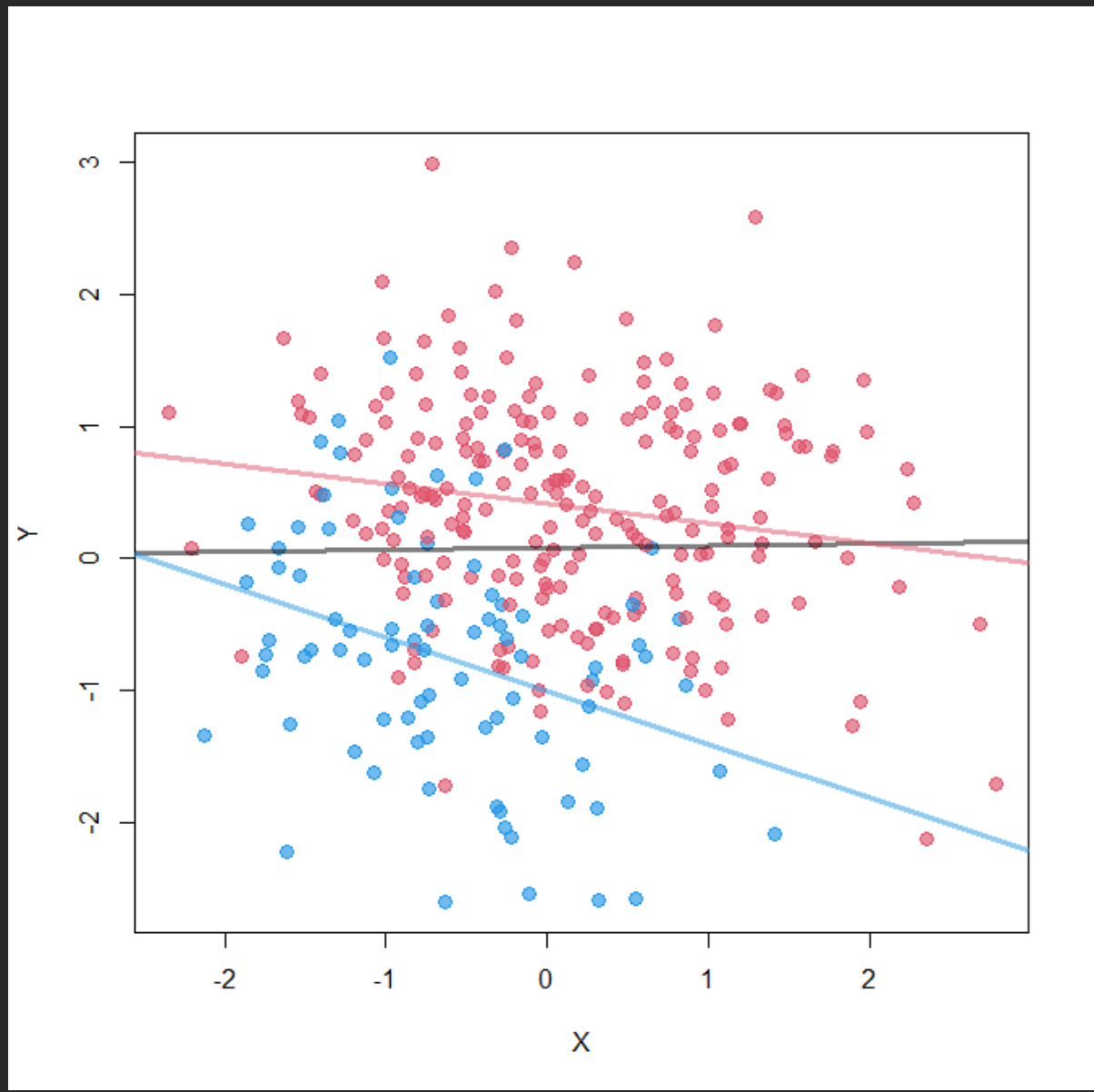
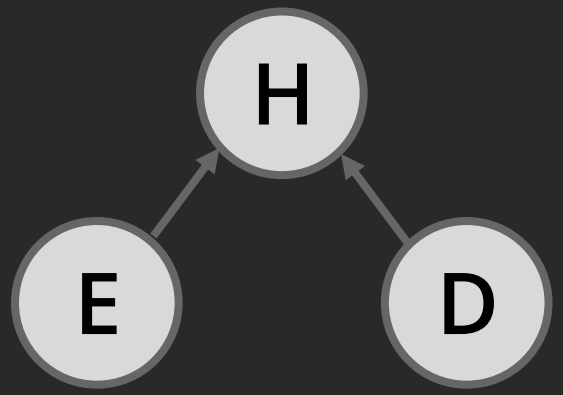



```
lm(formula = D ~ E + H)

            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.87400    0.09826  -8.894  < 2e-16 ***
E           -0.20154    0.05363  -3.758 0.000206 ***
H            1.29728    0.11625  11.160  < 2e-16 ***
```

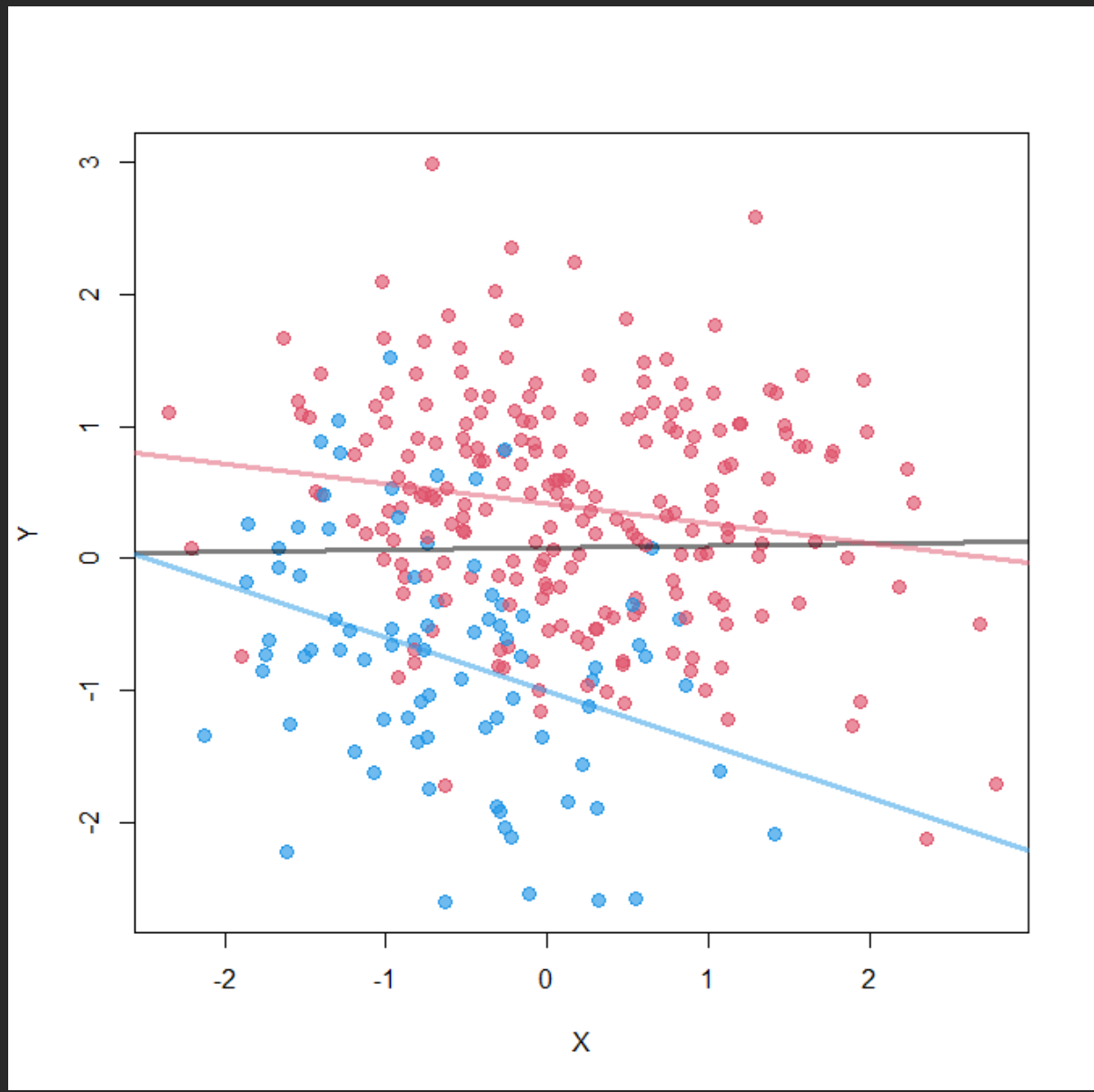


Collider: common effect

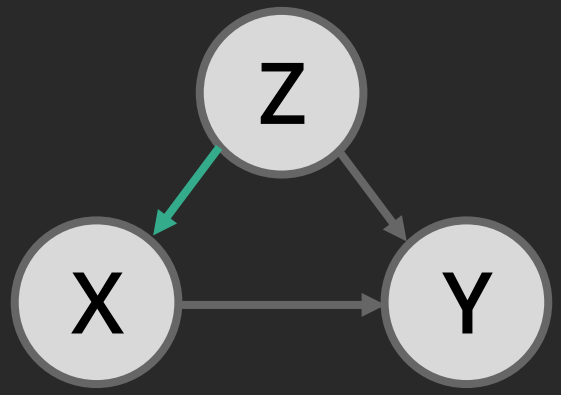


Sampling bias /
Bad (endogenous) controls

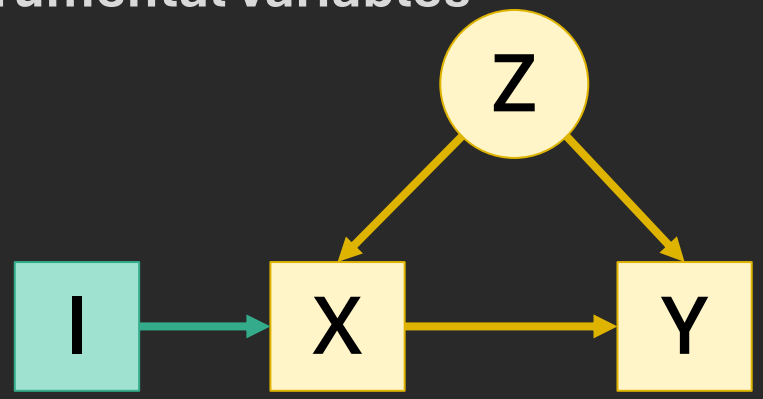
	Estimate	Std. Error	t value	Pr(>|t|)	
(Intercept)	-0.87400	0.09826	-8.894	< 2e-16	***
E	-0.20154	0.05363	-3.758	0.000206	***
H	1.29728	0.11625	11.160	< 2e-16	***



Alternatives to Backdoor Adjustments

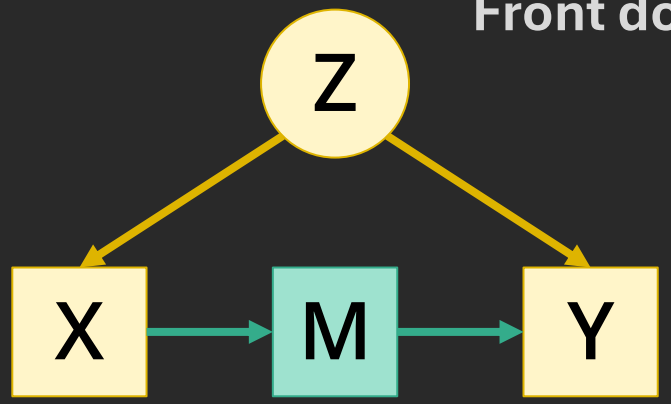


Instrumental variables



A 3rd variable (I) that only affects Y through its effect on X.
(e.g. Tax rate on smoking products)

Front door adjustment



A 3rd variable (M) that mediates the effect of X on Y.
(e.g. Tar levels in lungs mediating the effect of smoking on lung cancer)

Major takeaways of Causal Inference

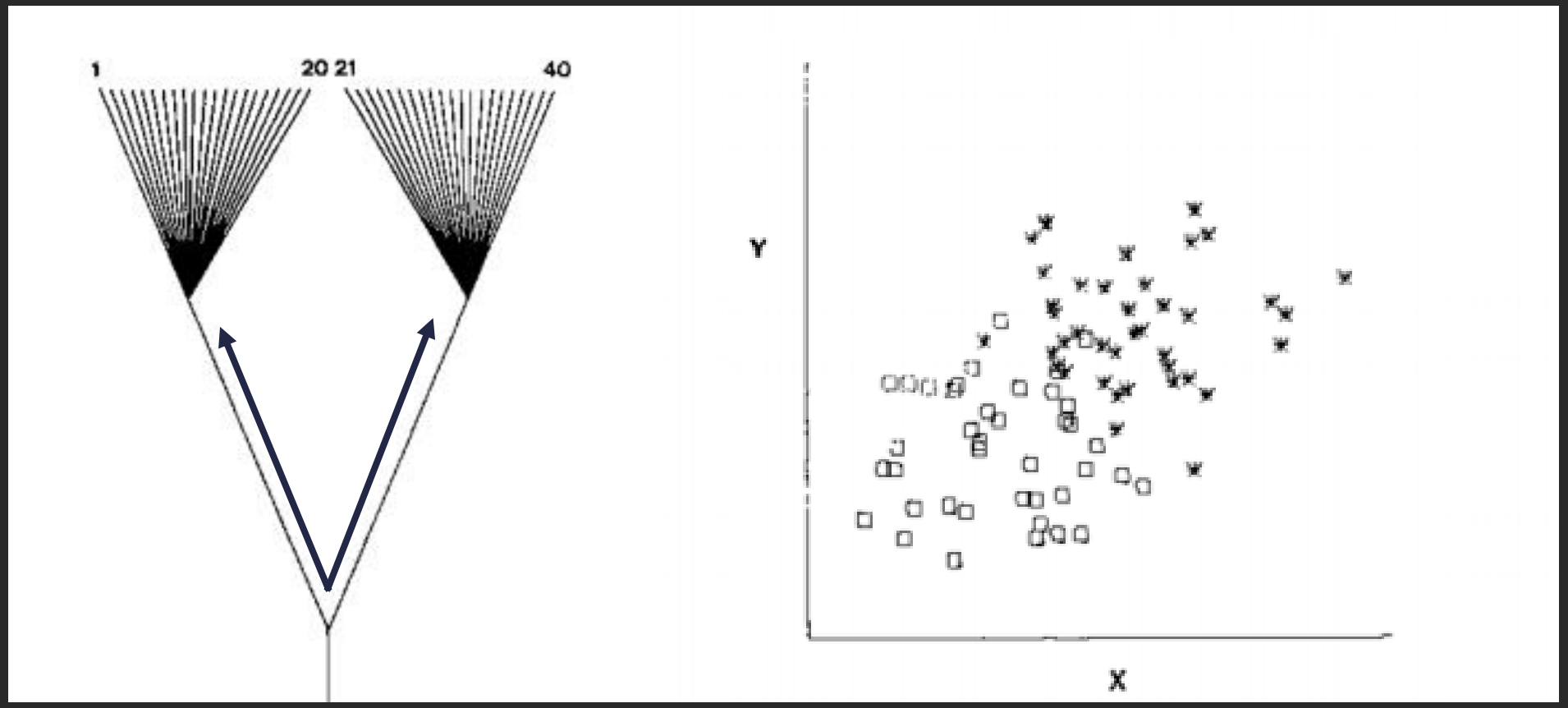
Prediction vs. Causal Inference

- Including all covariates that improve prediction (e.g. AIC model selection)
- Including covariates justified by the causal hypothesis

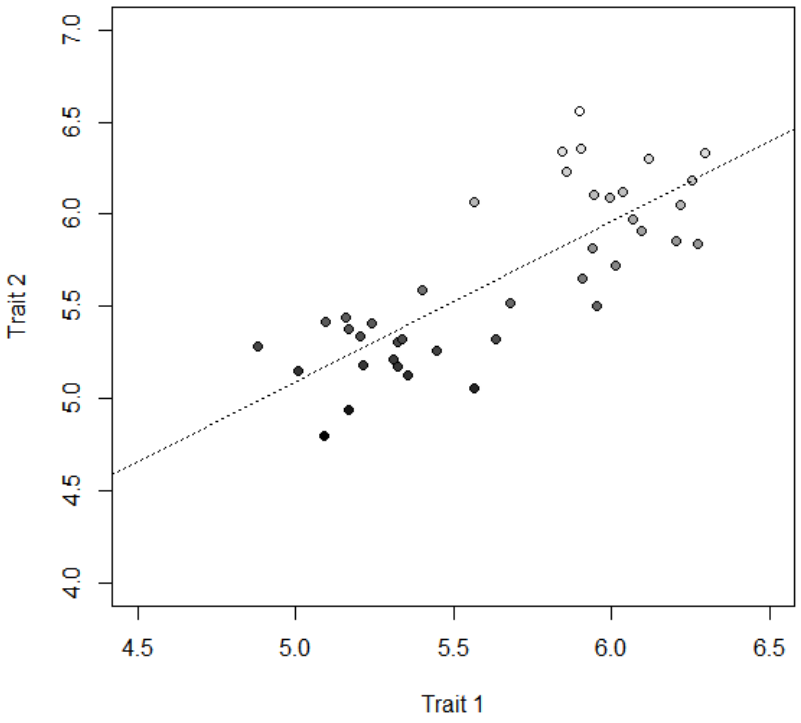
What we can do:

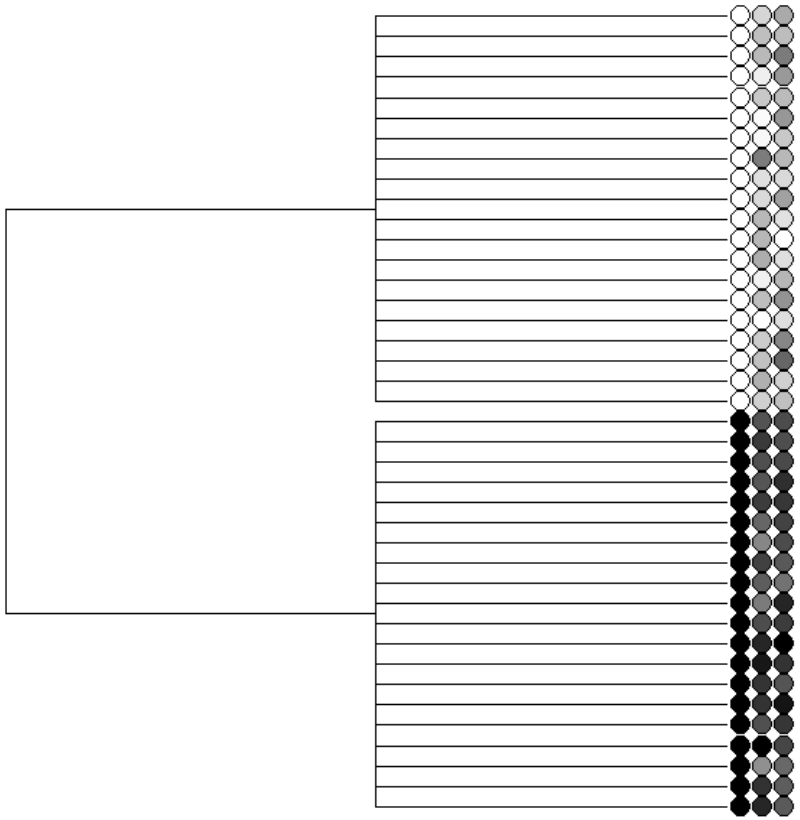
- DAGs are hypotheses of generating processes
- Makes causal assumptions explicit & justifies estimation/identification strategy *if the DAG were correct* (e.g. Backdoor adjustment)
- Doesn't solve "cor ≠ cause", but CAN REJECT implausible causal hypotheses
- Leaves *plausible* causal hypotheses & how they could be distinguished
- More identification strategies available than just Backdoor adjustment (e.g. frontdoor adjustment, instrumental variables)

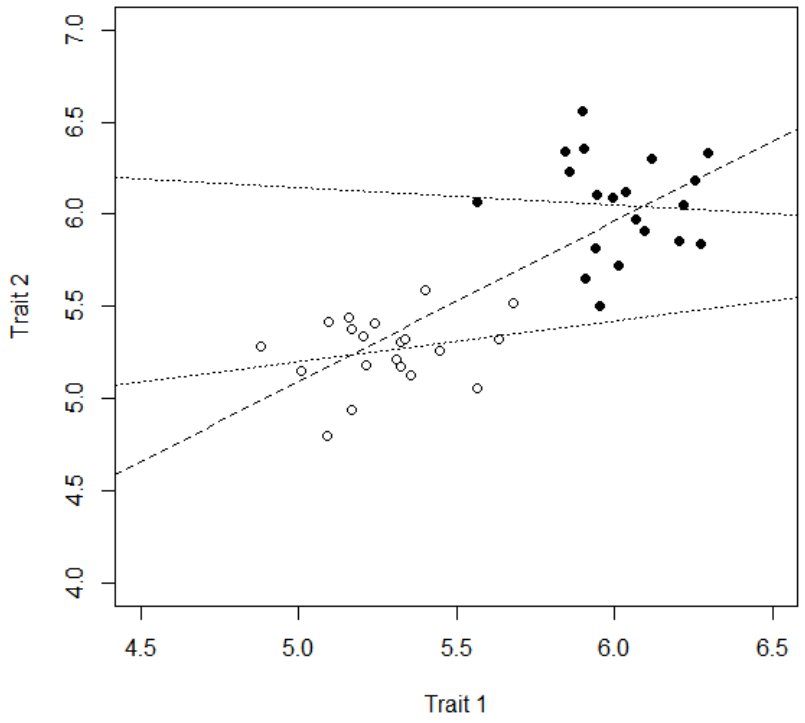
Application to PCM

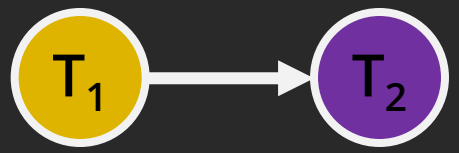


The Felsenstein Scenario

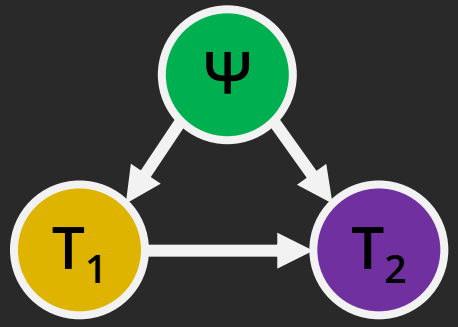




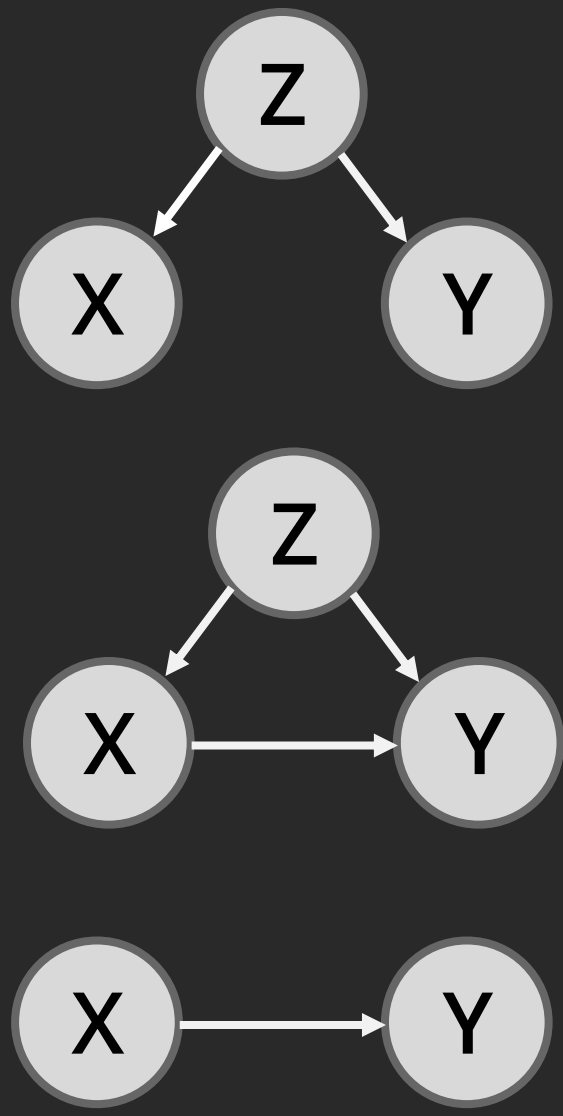


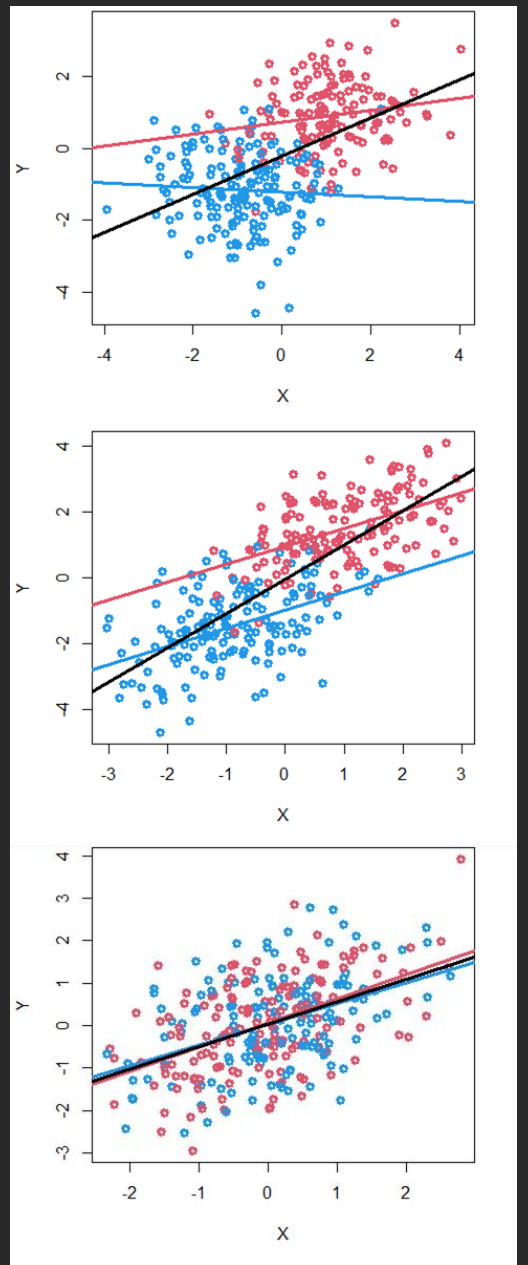


Spurious correlation due to confounding effect of phylogeny
(endogenous causal variable)



d-separation: testing the DAG





DAG implies X ⊥ Y

fit: Y ~ X + Z

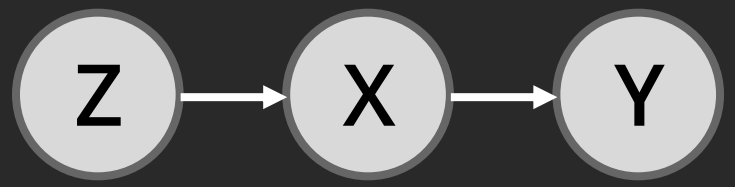
Testing models by a D-sep test.

Which model best predicts conditional independencies in the data?

Phylopath

Allows evaluation of d-separation among a specific set of causal models

Test for Conditional independence of: $Y|X \perp Z$

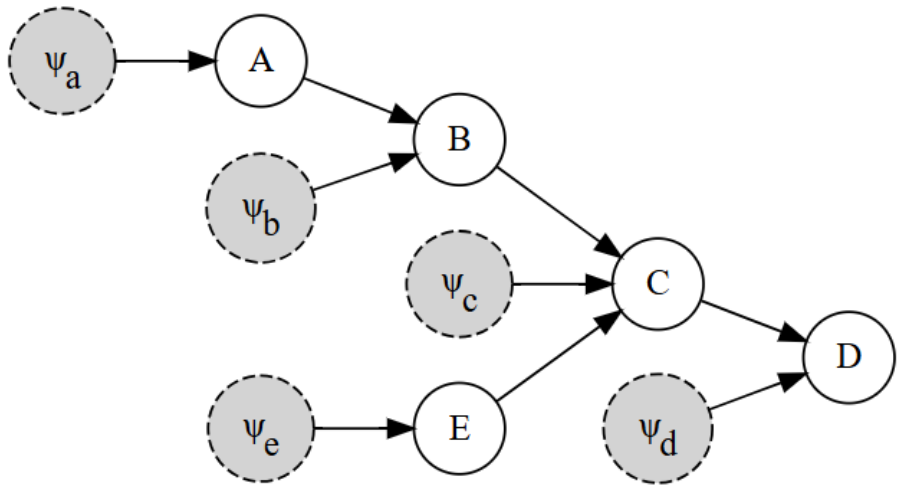


$Y|X \sim Z$ using PGLS (e.g. BM, BM + λ, OU)

If DAG is plausible, coefficient not significant

Accounts for expected phylogenetic covariance of residuals under model

Strong Assumptions:
- PGLS: only present-day values matter for causation (no causal lags)
- Residual covariance is only noise, not causal signal
- PGLS-based: historical effects
- Both DAG and residual model are correct
  e.g.: large shift violates BM, which will reject the DAG even if correct.



(Hardenberg & Gonzalez-Voyer 2013;Gonzalez-Voyer & Hardenberg 2014; van der Bijl 2018)

Omitted Variable Bias

So far, we've considered if confounder is measured.

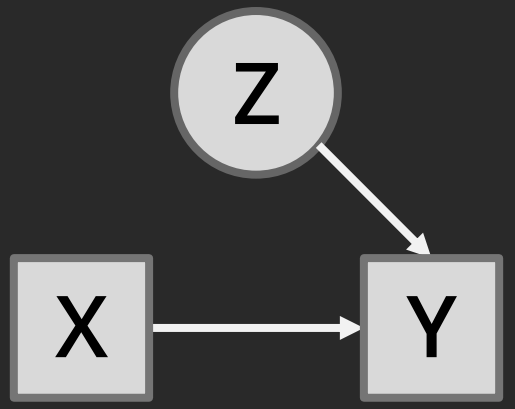
What if it is not known and/or measured?

Modeling correlation in error structure based on phylogeny = phylogenetic random effects

Random effects assumption: Random effects must not be correlated with any covariates in the regression; or else estimates are biased.

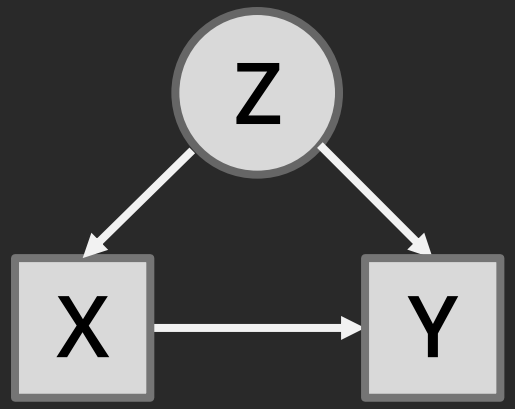
Observed

Omitted



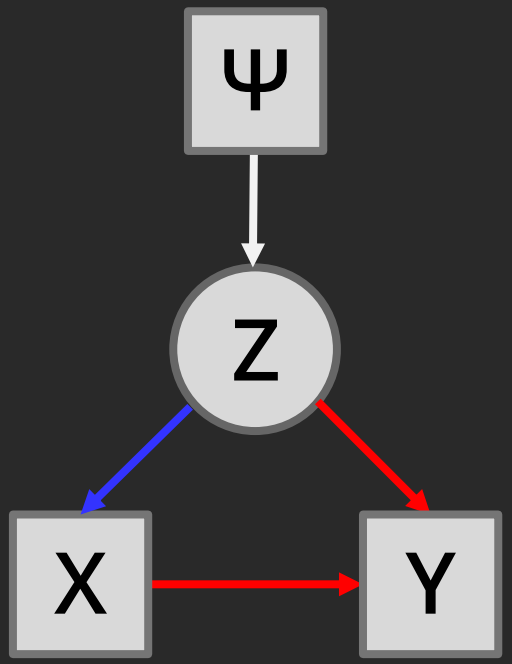
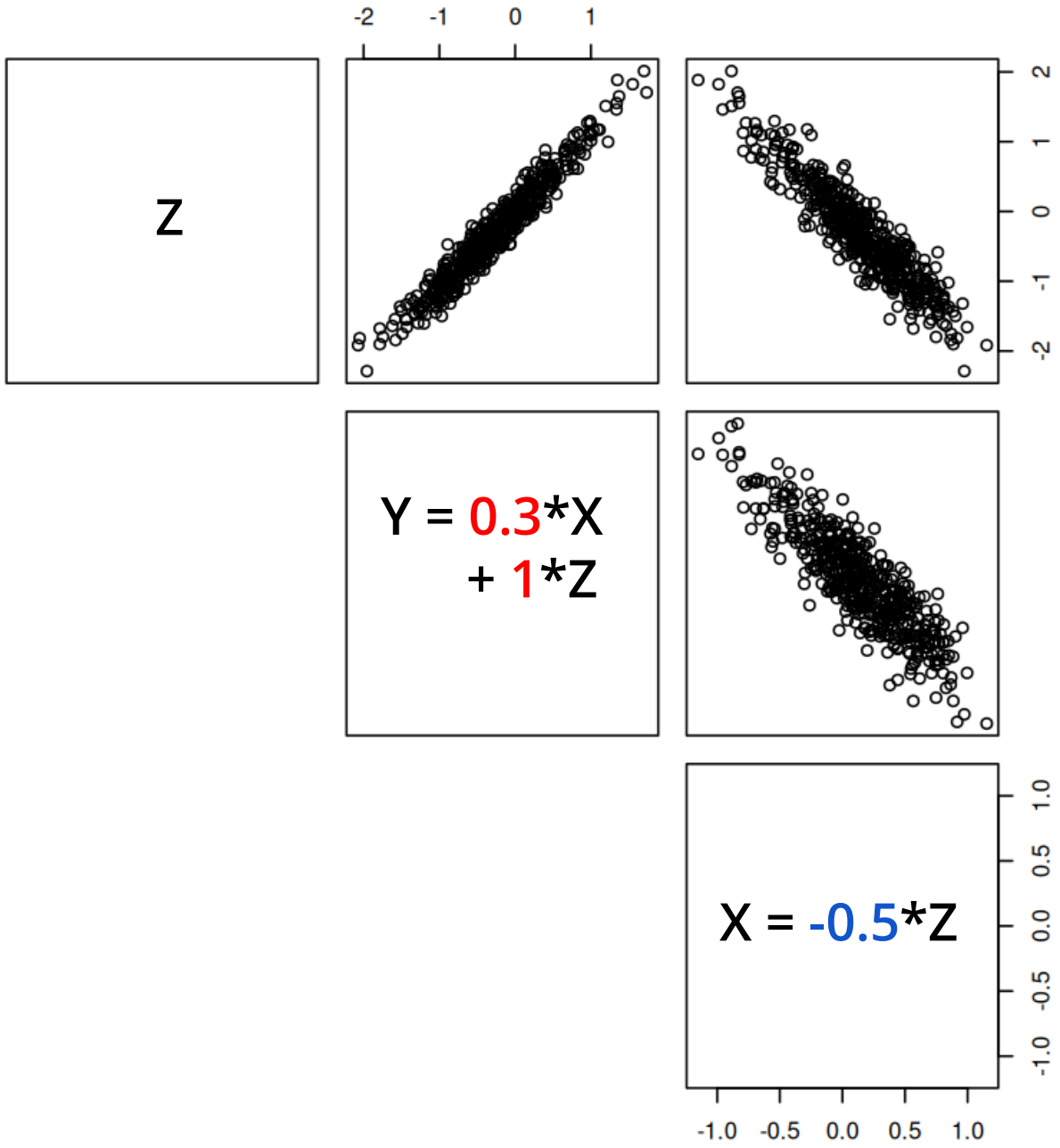
PGLS

(Assumption of a PGLS that would identify causal effect of X on Y)



PGLS w/endogeneity

(Estimate of causal effect of X on Y **biased!** Covariance estimator)

Observed

Omitted